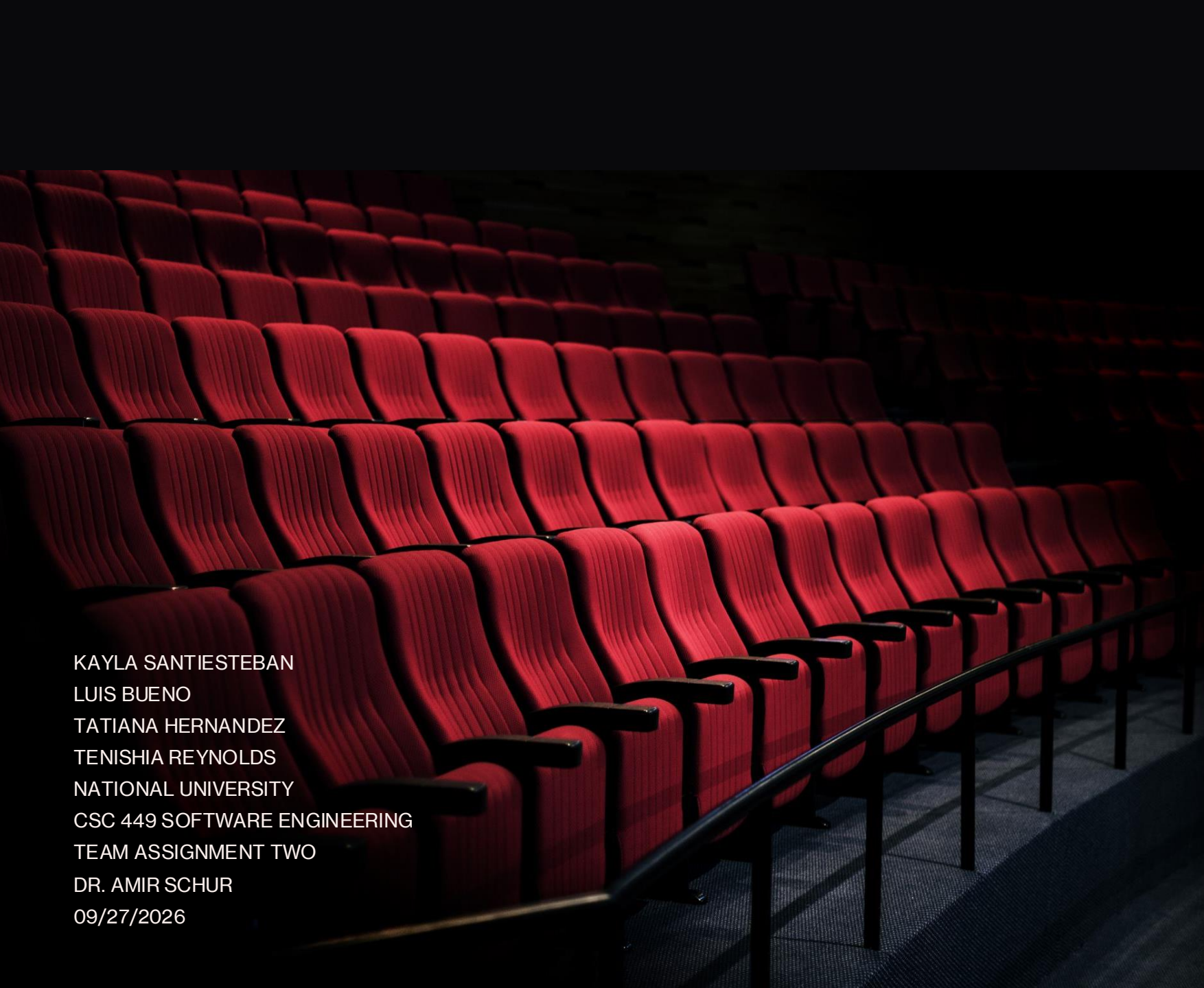




Movie Theater Seat Reservation System

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Software Architecture
Design & UML Views

Architecture Views

Reel Solutions focused on developing three key architectural views for the system during this sprint: the Use Case View, the Logical View, and the Process View. Together, these views helped define the system's required functionality, outline its structural design, and model its runtime behavior.

Use Case View: Customer

- Shows customer interactions with the system

Use Case View: Theater Manager

- Shows manager interactions with the system

Logical View

- Shows the system's major classes and relationships

Process View

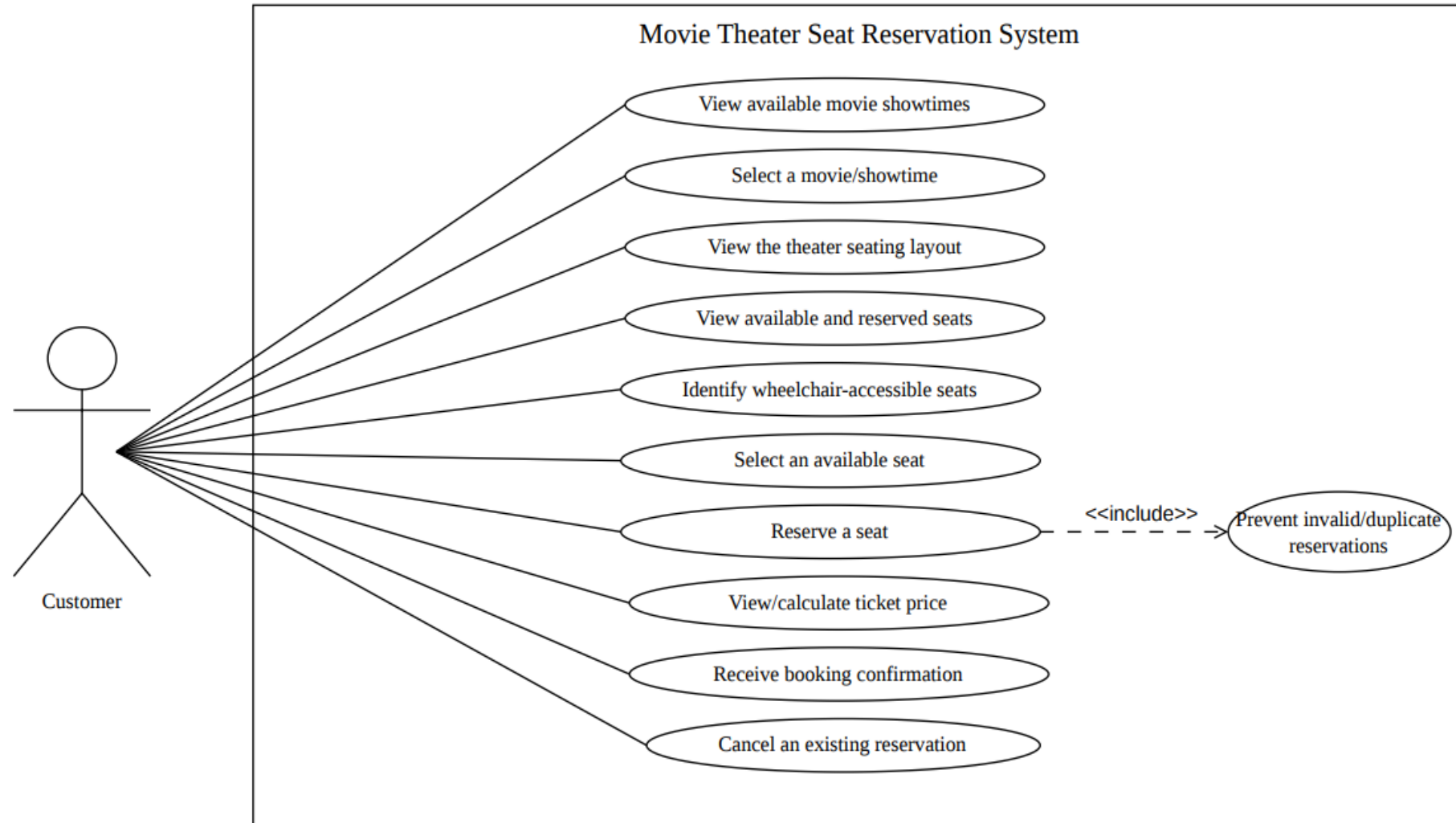
- Shows the seat-reservation workflow

Customer Use Case View

Actor: Customer

The Customer Use Case View shows how the customer interacts with the Movie Theater Seat Reservation System, including viewing showtimes, exploring the seating layout, selecting an available seat, completing a reservation, and managing their booking.

Customer Use Case View



Customer Use Case View

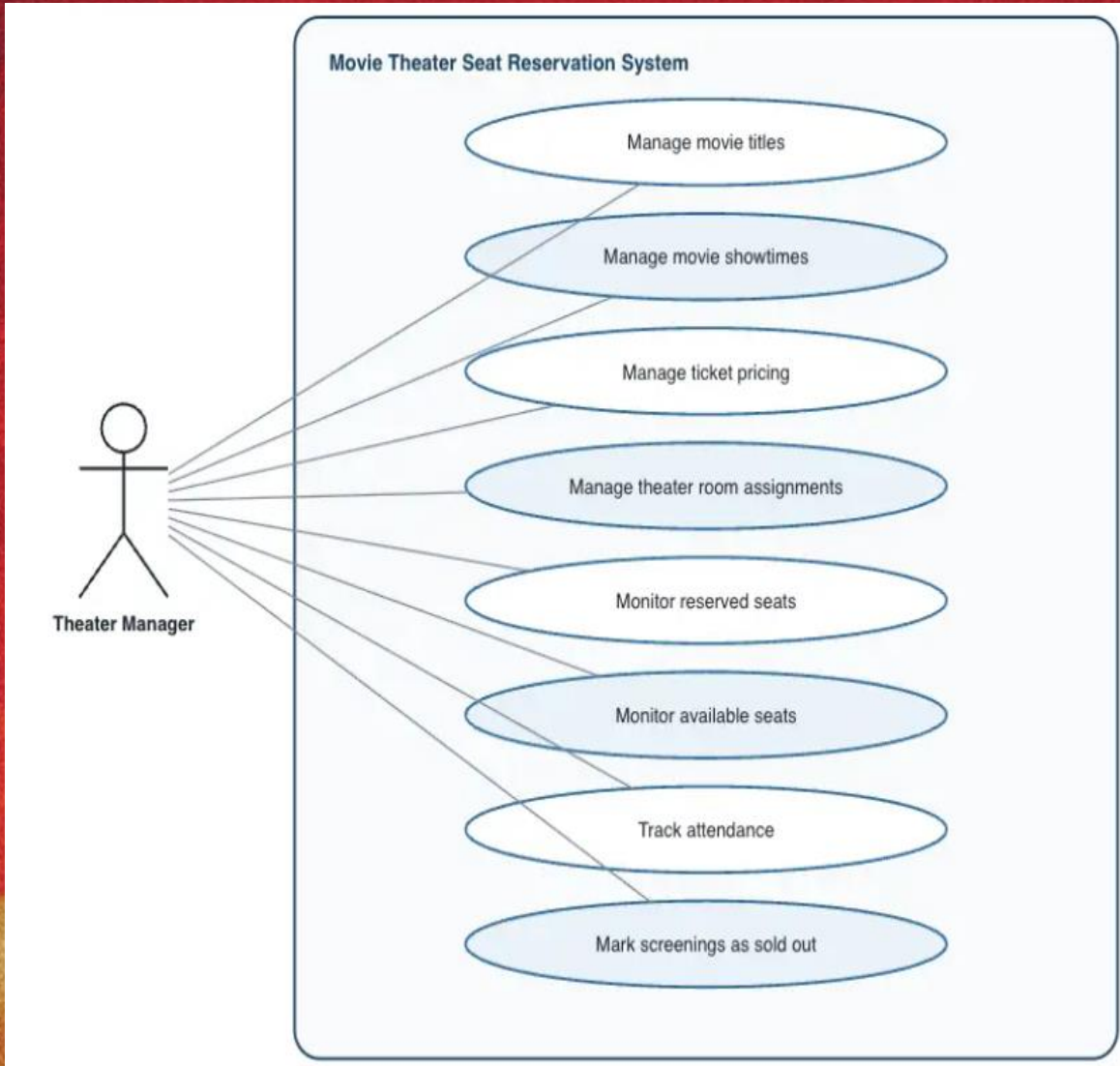
Use Case	Expected System Behavior or Outcome
View available movie showtimes	Display all upcoming movie showtimes for customer selection
Select a movie/showtime	Load the selected movie's screening details
View the theater seating layout	Show the seating map for the chosen showtime
View available and reserved seats	Clearly indicate which seats are open or already taken
Identify wheelchair-accessible seats	Highlight accessible seating options within the layout
Select an available seat	Allow the customer to choose any seat that is not reserved
Reserve a seat	Create a reservation for the selected seat and showtime
View/calculate ticket price	Display the total ticket cost before confirmation
Receive booking confirmation	Provide a confirmation message/code after successful reservation
Cancel an existing reservation	Allow the customer to cancel a previously made reservation

Theater Manager Use Case View

Actor: Theater Manager

The Theater Manager Use Case View shows how the theater manager interacts with the Movie Theater Seat Reservation System, including managing movies, showtimes, ticket pricing, theater rooms, and monitoring seat availability for each screening.

Theater Manager Use Case View

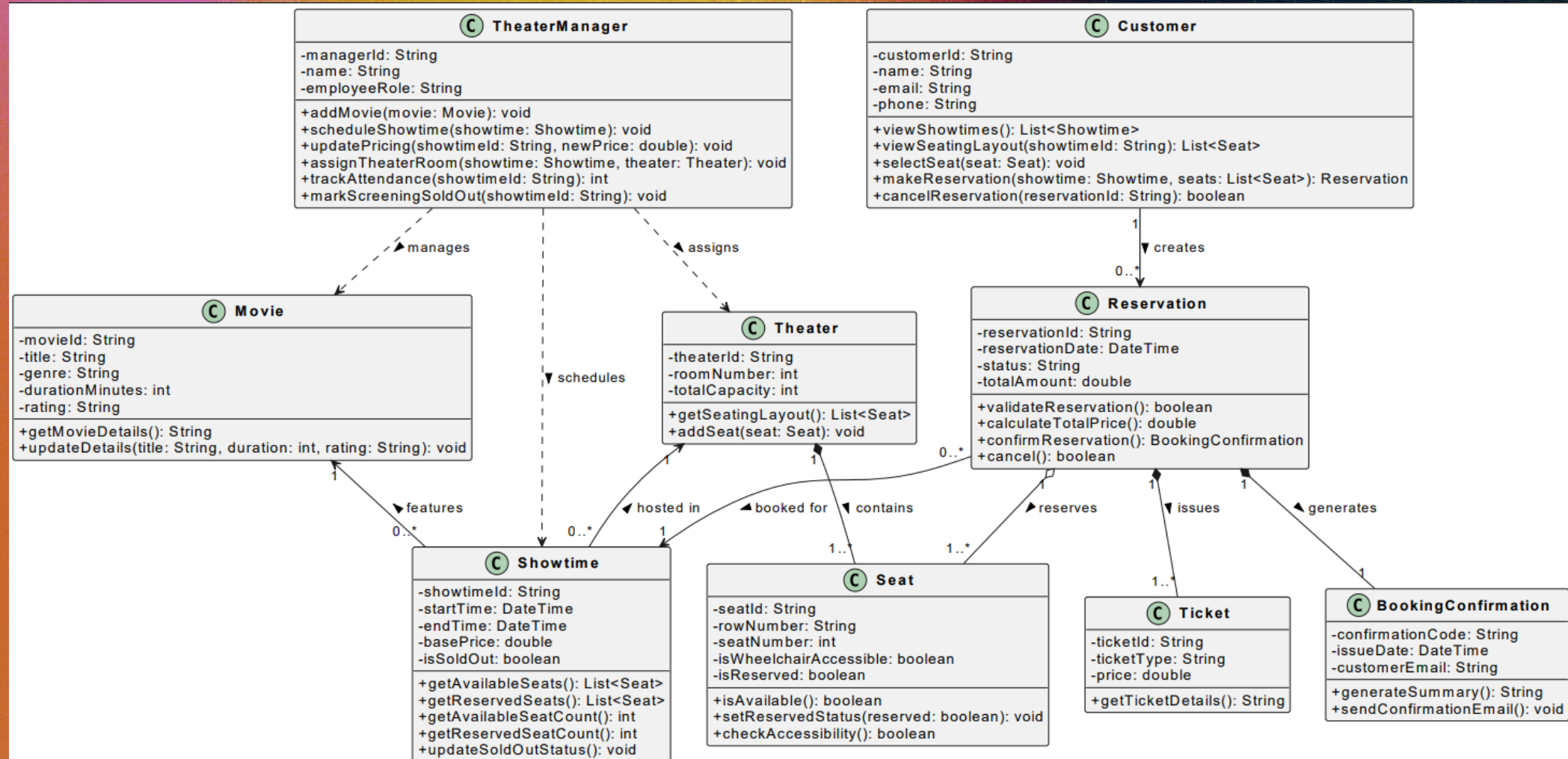


Use Case	Expected System Behavior or Outcome
Manage Movie Titles	Add, update, or remove movie titles in the theater catalog
Manage Movie Showtimes	create and maintain the scheduled screening times for each movie
Manage Ticket Pricing	set or update the ticket price associated with a screening
Manage Theater Room Assignments	assign each scheduled screening to the appropriate theater room
Monitor Reserved Seats	view how many seats have been reserved for a selected showing
Monitor Available Seats	view how many seats remain available for a selected showing
Track Attendance	use reserved and available seat totals to monitor expected attendance
Mark Screenings As Sold-out	change a screening to sold-out status when no additional seats are available

Logical View

Logical View shows the system's main classes, their attributes, methods, and relationships. This view illustrates how core components such as movies, showtimes, seats, reservations, and users connect and work together within the Movie Theater Seat Reservation System.

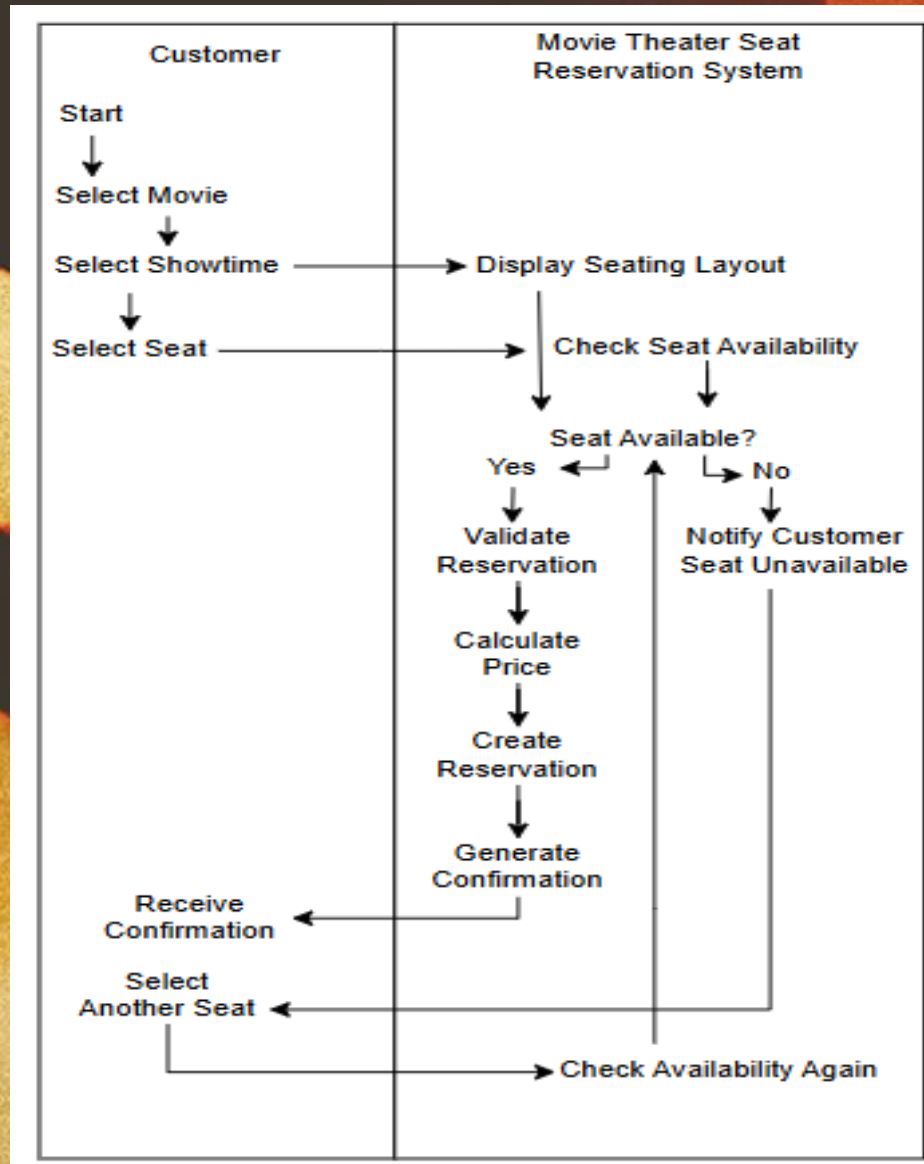
Logical View: UML Diagram



Process View

Process View shows the step-by-step flow of the seat-reservation process, including selecting a movie and showtime, viewing the seating layout, choosing a seat, validating availability, calculating pricing, creating the reservation, and generating the booking confirmation. This view also captures the system's response when a seat is already reserved.

Process View: Seat Reservation UML Diagram



System Architectures

Use Case View shows who uses the system and what they do.

Logical View shows the major classes and how they relate.

Process View shows how a customer reservation moves through the system.

Core Reservation Flow:

Movie → Showtime → Theater → Seat → Reservation → Ticket → Booking Confirmation